

8.6 Reflection



Core skills covered in this section:

- ▶ Reflection is a geometric transformation that maps a point to its mirror-image position across an axis or plane of symmetry.
- ▶ Constructing a reflection matrix using the normal vector.
- ▶ A reflection matrix is a symmetric matrix and also an orthogonal matrix.
- ▶ The inverse of a reflection matrix is itself.
- ▶ A reflection transformation preserves the area of a shape but flips its orientation, changing the sign of the determinant.
- ▶ The relationship between orthogonal projection matrices and reflection matrices, involving the Householder matrix.

Reflection in the Plane

Reflection (also sometimes called mirroring) maps a point to its mirror-image position across a given axis or plane of symmetry. Mathematically, a reflection transformation is usually carried out via matrix multiplication and touches on many linear algebra concepts.

Given a line through the origin whose normal vector is $\mathbf{n}$, using this line as the axis of symmetry, the matrix that performs the reflection is

$$\mathbf{M} = \mathbf{I} - 2 \cdot \frac{\mathbf{n} @ \mathbf{n}^T}{\|\mathbf{n}\|^2} = \mathbf{I} - 2 \cdot \frac{\mathbf{n} @ \mathbf{n}^T}{\mathbf{n}^T @ \mathbf{n}}$$

In particular, if the normal vector is the unit vector $\hat{\mathbf{n}}$, then $\mathbf{M}$ is

$$\mathbf{M} = \mathbf{I} - 2 \cdot \hat{\mathbf{n}} @ \hat{\mathbf{n}}^T$$

For example, when $a_1x_1 + a_2x_2 = b$ passes through the origin, $b = 0$, i.e., $a_1x_1 + a_2x_2 = 0$.

The (through-the-origin) line $a_1x_1 + a_2x_2 = 0$ has normal vector

$$\mathbf{n} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

The reflection matrix for reflecting about $a_1x_1 + a_2x_2 = 0$ is

$$\begin{aligned}
 \mathbf{M} &= \mathbf{I} - 2 \cdot \frac{\mathbf{n} @ \mathbf{n}^T}{\mathbf{n}^T @ \mathbf{n}} \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{a_1^2 + a_2^2} \times \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} @ \begin{bmatrix} a_1 & a_2 \end{bmatrix} \\
 &= \frac{1}{a_1^2 + a_2^2} \left(\begin{bmatrix} a_1^2 + a_2^2 & 0 \\ 0 & a_1^2 + a_2^2 \end{bmatrix} - \begin{bmatrix} 2a_1^2 & 2a_1a_2 \\ 2a_1a_2 & 2a_2^2 \end{bmatrix} \right) \\
 &= \frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 - a_1^2 & -2a_1a_2 \\ -2a_1a_2 & a_1^2 - a_2^2 \end{bmatrix}
 \end{aligned}$$

Simply choosing a unit normal vector lets us construct the reflection transformation for the (through-the-origin) hyperplane defined by that normal vector. The reflection matrix constructed from the normal vector applies not only in two and three dimensions but holds in any n -dimensional Euclidean space.

A Few Sets of Examples

Let's look at a few special examples below.

Figure 1 (a): reflecting about the horizontal axis x_1 . The equation of the horizontal axis is $x_2 = 0$, so it is easy to find that its unit normal vector is

$$\hat{\mathbf{n}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Compute its reflection matrix

$$\mathbf{M} = \mathbf{I} - 2 \cdot \hat{\mathbf{n}} @ \hat{\mathbf{n}}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} @ \begin{bmatrix} 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The vector $[x_1, x_2]^T$'s reflection about the horizontal axis is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}$$

This agrees with Figure 1 (a): the x_1 coordinate stays unchanged, and the x_2 coordinate flips sign.

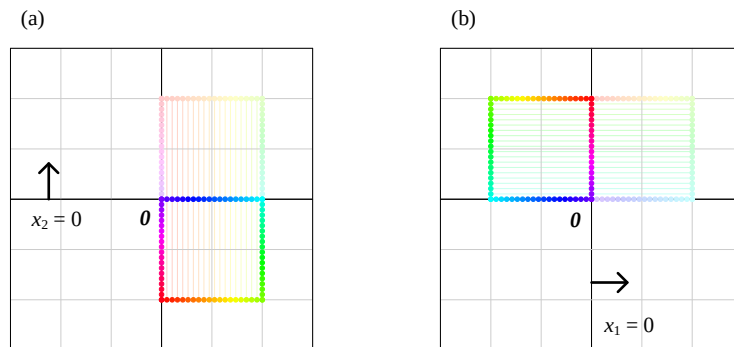


Figure 1. Symmetric About the Horizontal and Vertical Axes

Figure 1 (b): reflecting about the vertical axis x_2 . The equation of the vertical axis is $x_1 = 0$, and its unit normal vector is

$$\hat{n} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Compute the corresponding reflection matrix

$$\mathbf{M} = \mathbf{I} - 2 \cdot \hat{n} @ \hat{n}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \end{bmatrix}^T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

The vector $[x_1, x_2]^T$'s reflection about the vertical axis is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_1 \\ x_2 \end{bmatrix}$$

This agrees with the observation from Figure 1 (b): the x_1 coordinate flips sign, and the x_2 coordinate stays unchanged.

Let's look at the two examples in Figure 2.

Figure 2 (a): the mirror line through the origin is $x_1 - x_2 = 0$, and the corresponding reflection matrix is

$$\mathbf{M} = \mathbf{I} - 2 \cdot \hat{n} @ \hat{n}^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



Please manually verify $\mathbf{M}$.

The vector $[x_1, x_2]^T$'s reflection about $x_1 - x_2 = 0$ is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

The equation above is equivalent to swapping the horizontal and vertical coordinates.

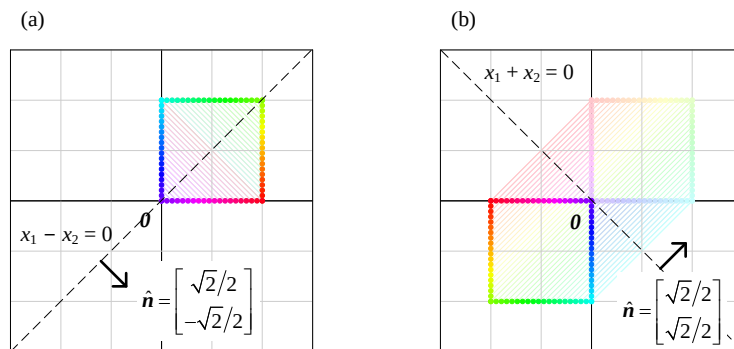


Figure 2. Symmetric About the 45° and 135° Lines Through the Origin

Figure 2 (b): the mirror line through the origin is $x_1 + x_2 = 0$, and the corresponding reflection matrix is

$$\mathbf{M} = \mathbf{I} - 2 \cdot \hat{n} @ \hat{n}^T = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

? Please compute in $\mathbf{M}$.

The vector $[x_1, x_2]^T$'s reflection about $x_1 + x_2 = 0$ is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ -x_1 \end{bmatrix}$$

The equation above is equivalent to swapping the horizontal and vertical coordinates and negating them.

Matrix Properties

Clearly, in , the square matrix $\mathbf{M}$ is a symmetric matrix, i.e., it satisfies

$$\mathbf{M} = \mathbf{M}^T$$

The reflection matrix $\mathbf{M}$ is also an orthogonal matrix, i.e., it satisfies

$$\mathbf{M}\mathbf{M}^T = \mathbf{M}^T\mathbf{M} = \mathbf{I}$$

This means that the inverse of the reflection operation is itself. It also means that the column vectors of $\mathbf{M}$ form an orthonormal basis.

Combining and , we find that

$$\mathbf{M}^2 = \mathbf{I}$$

This means that applying the same reflection twice in succession restores the original shape, as shown in the Figure 3 example.

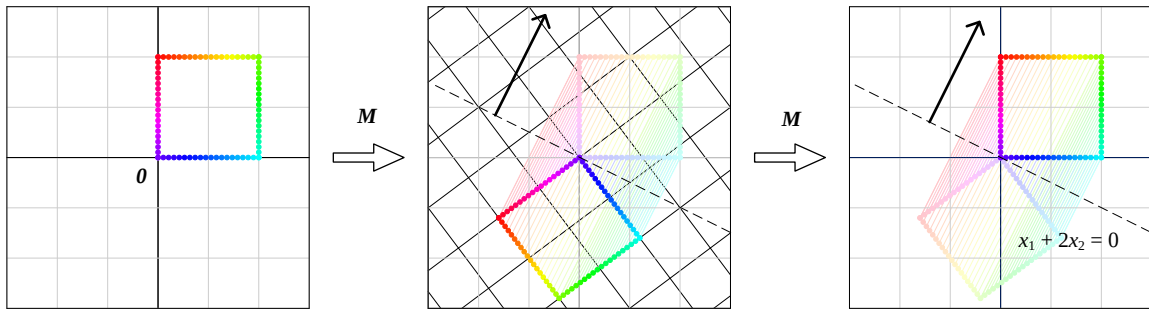


Figure 3. Applying the Same Reflection Twice Restores the Original Shape

Compute the determinant of

$$\det(\mathbf{M}) = \det \left(\frac{-1}{a_1^2 + a_2^2} \begin{bmatrix} a_1^2 - a_2^2 & 2a_1a_2 \\ 2a_1a_2 & a_2^2 - a_1^2 \end{bmatrix} \right) = \frac{(a_1^2 - a_2^2)(a_2^2 - a_1^2) - 4a_1^2a_2^2}{(a_1^2 + a_2^2)^2} = \frac{-(a_1^2 + a_2^2)^2}{(a_1^2 + a_2^2)^2} = -1$$

This book mentioned earlier that the sign of the determinant relates to the orientation of the geometric operation. Simply put, if the determinant is positive, the transformation preserves the order of the basis vectors; if the determinant is negative, it reverses the order of the basis vectors — as with reflection.

Although reflection does not change the area of a shape, the image is flipped and the order of the column vectors is swapped.

Deriving $\det(kA)$ uses a property covered in the determinants chapter. For an $n \times n$ square matrix A , the determinant of the scalar multiple kA is

$$\det(kA) = k^n \det(A)$$



Please review the material on determinant properties covered earlier in this book.

The Relationship Between Orthogonal Projection and Reflection

in the equation above, the matrix M has its own name: the Householder matrix. The Householder matrix is directly related to orthogonal projection — let's discuss this connection below.

As Figure 4 shows, the vector a and the vector b are mirror-symmetric about the line through the origin whose normal vector is n .

And the difference between the two, $a - b$, is twice the vector projection onto the n direction, i.e.,

$$a - b = 2 \text{proj}_n a$$

The equation above can be rearranged to give b ; this is left to interested readers.

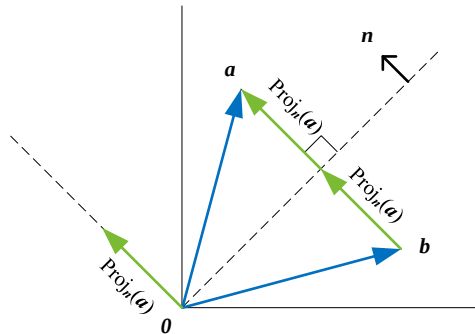


Figure 4. The Householder Matrix and Orthogonal Projection

Reflection in Three-Dimensional Space

this reflection matrix also applies to reflection in the two-dimensional plane. As mentioned earlier, for a plane, its normal vector is the vector perpendicular to the plane.

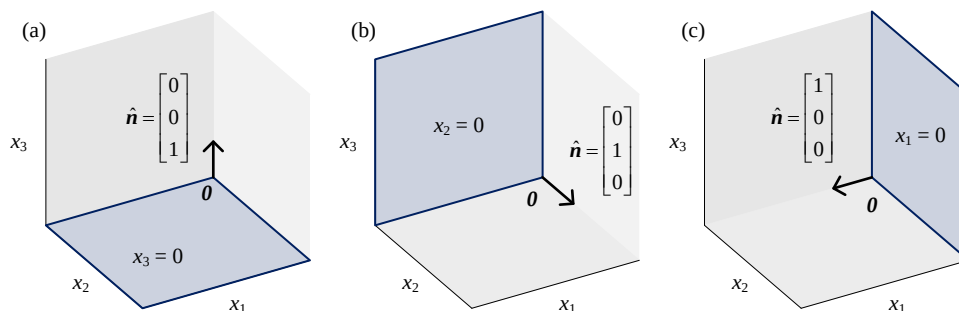


Figure 5. The Normal Vector of a Plane

As Figure 5 (a) shows, the unit normal vector of the x_1x_2 plane is $[0, 0, 1]^T$, and the reflection matrix is

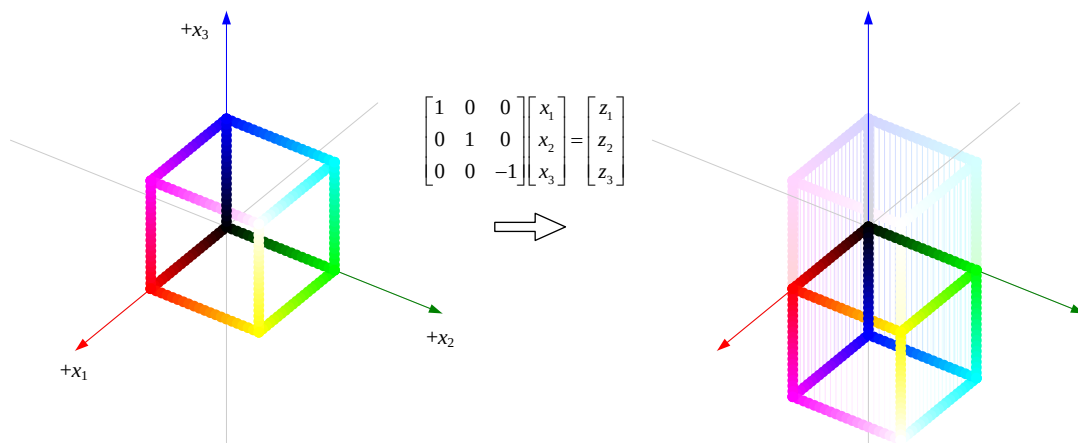
$$\mathbf{M} = \mathbf{I} - 2 \cdot \hat{\mathbf{n}} @ \hat{\mathbf{n}}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \times \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

? Please compute the determinant of the reflection matrix above.

As Figure 6 shows, the result of reflecting the three-dimensional column vector $\mathbf{x}$ about the x_1x_2 plane is

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ -x_3 \end{bmatrix}$$

That is, the x_1, x_2 coordinates stay unchanged, and the x_3 coordinate flips sign.

Figure 6. Reflection About the x_1x_2 Plane, Three-Dimensional Space

Please also use Figure 7 to review, from earlier in this book, how the right-hand rule can be used to determine the sign of the determinant of .

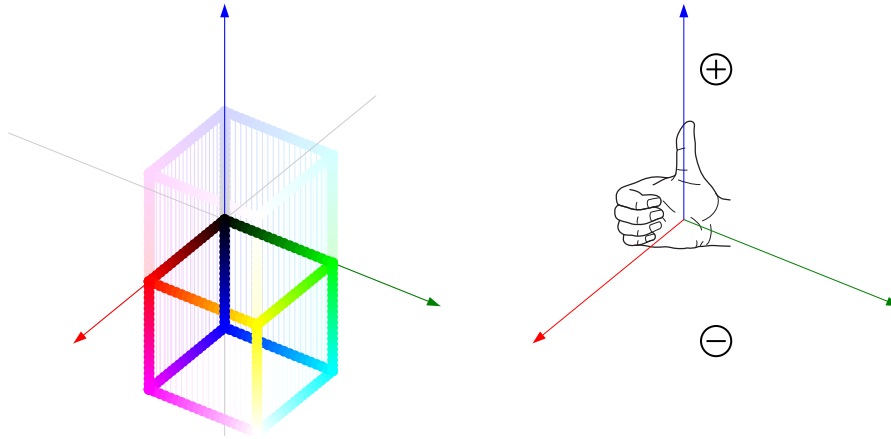
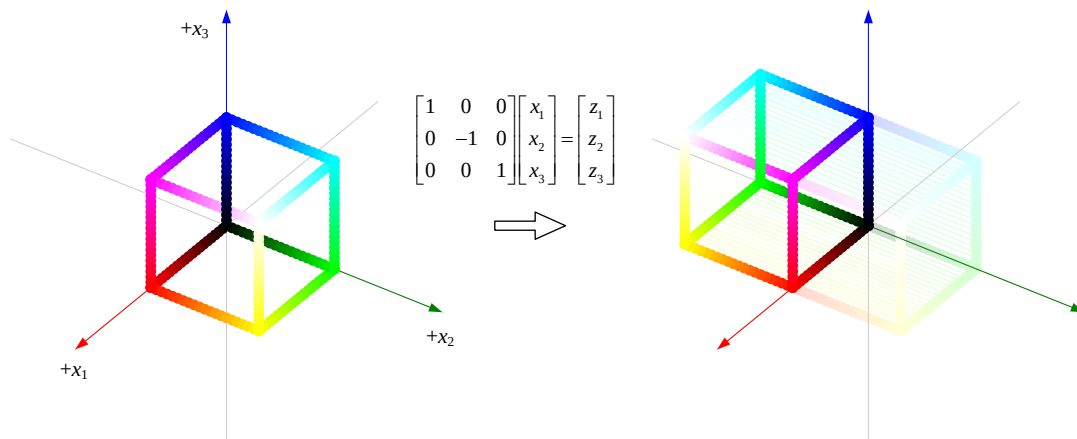


Figure 7. Using the Right-Hand Rule to Determine the Sign of the Determinant

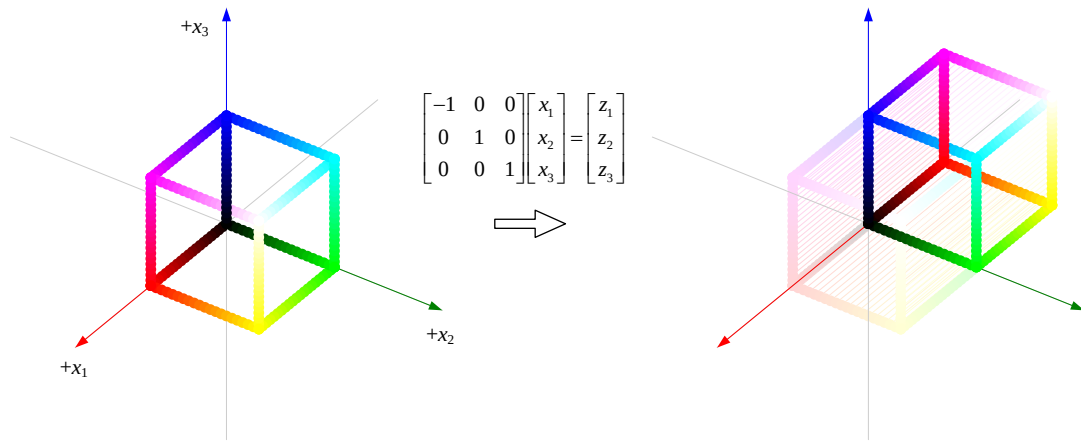
As Figure 5 (b) shows, the unit normal vector of the x_1x_3 plane is $[0, 1, 0]^T$, and the reflection matrix is

$$M = I - 2 \cdot \hat{n} @ \hat{n}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \times \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

As Figure 8 shows, the x_1, x_3 coordinates stay unchanged, and the x_2 coordinate flips sign.

Figure 8. Reflection About the x_1x_3 Plane, Three-Dimensional Space

? Please compute on your own the reflection matrix corresponding to Figure 9, and state the resulting coordinate transformation.

Figure 9. Reflection About the x_2x_3 Plane, Three-Dimensional Space

Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. For each of the following lines through the origin in the plane, compute its normal vector and reflection matrix.

- ▶ $-x_2 = 0$
- ▶ $-x_1 = 0$
- ▶ $x_1 - 2x_2 = 0$
- ▶ $2x_1 - x_2 = 0$
- ▶ $x_1 + 2x_2 = 0$
- ▶ $2x_1 + x_2 = 0$

Q2. For each of the following planes through the origin in three-dimensional space, compute the normal vector and the reflection matrix.

- ▶ $x_2 = 0$
- ▶ $x_1 = 0$
- ▶ $x_1 + x_2 = 0$
- ▶ $x_1 + x_3 = 0$
- ▶ $x_2 + x_3 = 0$
- ▶ $x_1 + x_2 + x_3 = 0$
- ▶ $x_1 - x_2 - x_3 = 0$

Q3. Please modify LA_08_02_01.ipynb to visualize the planar reflection operations in Figure 1 and Figure 2.

Q4. Please modify LA_08_03_01.ipynb to visualize the three-dimensional reflection operations in Figure 6, Figure 8, and Figure 9.

Q5. For each of the following hyperplane normal vectors, compute the reflection matrix.

► $[0, 0, 0, 1]^T$

► $[0, 0, 1, 0]^T$

► $[1, 1, 1, 1]^T$

Q6. For each of the following hyperplanes, compute the reflection matrix.

► $x_1 + x_2 + x_3 + x_4 = 0$

► $x_1 - x_2 - x_3 + x_4 = 0$